

UNIVERSIDAD DE COSTA RICA
SISTEMA DE ESTUDIOS DE POSGRADO

**FENOLOGÍA DE ABEJAS EUGLOSSINAS Y SU INTERACCIÓN CON LAS
ORQUÍDEAS EN UN BOSQUE HUMEDO TROPICAL DE COSTA RICA**

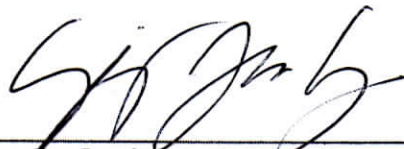
Tesis sometida a la consideración de la Comisión del Programa de Estudios de
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DEDICATORIA

A mi familia y amigos,
que siempre estuvieron presentes

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RESUMEN GENERAL

La polinización es una interacción clave para la biodiversidad y los ecosistemas. En los trópicos, las abejas Euglossini desempeñan un papel crucial en la polinización de las orquídeas, pero se sabe poco sobre cómo los cambios climáticos, como las precipitaciones, afectan a estas interacciones. Aunque las abejas dependen de los recursos florales, la dinámica de estas relaciones varía con el tiempo. La fenología estudia los ciclos biológicos y sus relaciones con el clima, que afectan a la abundancia de insectos a lo largo del año. Este estudio analiza cómo interactúan las abejas Euglossini y las orquídeas en un bosque tropical húmedo de Costa Rica y cómo influye la precipitación en esas interacciones. La investigación se llevó a cabo en la Estación Biológica La Selva, Costa Rica, que se caracteriza por un clima tropical húmedo. Se recogieron muestras mensuales de machos de abejas Euglossini mediante cebos químicos. Se contaron todos los machos y se colectaron aquellos que tenían polinarios en sus cuerpos, para construir redes de interacción bipartitas y ver el efecto de la precipitación en la riqueza, abundancia de los machos y los cambios en las redes de interacción entre las temporadas de bajas y altas precipitaciones. Un total de 4994 machos fueron contados y fueron capturados 623 machos que contenían 643 polinarios. Se encontraron patrones de estacionalidad en 20 especies de euglosinos, y 4 mostraron correlaciones significativas con las precipitaciones. La red durante la estación lluviosa fue relativamente grande, pero ambas estaciones tuvieron valores similares de anidamiento. Hubo una correlación positiva entre la abundancia de machos euglosinos y polinarios. Este estudio es pionero en documentar la fenología de las interacciones entre las abejas Euglossini y las orquídeas en la región Neotropical, destacando el impacto de la precipitación en estas redes ecológicas. A diferencia de estudios anteriores en los que las interacciones se concentraban en las estaciones secas, se observó una mayor actividad durante la estación lluviosa. Futuros estudios deberán integrar herramientas moleculares y análisis químicos para mejorar la identificación de especies y comprender mejor las relaciones entre las abejas Euglossini y las orquídeas.

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INTRODUCCIÓN GENERAL:

Las interacciones biológicas son un componente fundamental de los ecosistemas, ya que sustentan la vida y su ausencia puede conducir a la pérdida de biodiversidad (Jordano, 2016a). Estas interacciones no solo reflejan la dinámica y calidad de los ecosistemas, sino que también permiten estimar la biodiversidad de manera más precisa, por medio de los grupos funcionales (Jordano, 2016b, 2016a). Un ecosistema saludable se caracteriza por una diversidad elevada de interacciones, tanto entre miembros de la misma especie (intraespecíficas) como entre especies diferentes (interespecíficas), las cuales incluyen relaciones como el comensalismo, amensalismo, depredación, parasitismo y mutualismo. Muchas de estas interacciones son esenciales para el funcionamiento de los ecosistemas y proporcionan servicios ecosistémicos clave para los seres humanos, como la polinización (Mommott et al., 2007; Theodorou, 2022; Valiente-Banuet et al., 2015).

La polinización es el proceso de transferencia del polen desde el estambre de una flor hasta el estigma receptivo de otra o la misma flor, permitiendo luego la fertilización. Este proceso puede realizarse a través de distintos medios, como el agua (hidrofilia), el viento (anemofilia) o mediante la intervención de organismos vectores, como aves, mamíferos e insectos, conocidos como polinizadores (Faegri & Pijl, 2013; Mayer et al., 2004; Ricketts et al., 2008; Torezan-Silingardi et al., 2021). Las plantas con flores han desarrollado diversas estrategias para facilitar su reproducción y recompensas para atraer a los polinizadores, utilizando una amplia gama de colores, formas y tamaños en sus órganos florales (Nadot & Carrive, 2021; Schiestl & Johnson, 2013; Whitney & Glover, 2007). Algunas especies recurren a mecanismos de engaño, como el engaño alimenticio, sexual o la imitación de sitios de oviposición, como ocurre en familias como Asclepiadaceae, Caricaceae, Cucurbitaceae y Begoniaceae (Castillo et al., 1999; Ferdy et al., 1998; Jersáková et al., 2006). Otros linajes, en cambio, ofrecen recompensas como polen, néctar, resinas, aromas o aceites florales para asegurar visitas constantes de los polinizadores, una estrategia observada en taxones como Malpighiaceae y Passifloraceae (Renner, 2021; Shivanna & Tandon, 2014; Zaríman et al., 2022). Dentro de la familia Orchidaceae, existe una notable diversidad de estrategias de polinización, lo que resalta su adaptabilidad y especialización en este proceso (Karremans, 2023; Ackerman et al., 2023).

Generalidades de las orquídeas:

Existen aproximadamente 30,000 especies de orquídeas distribuidas globalmente, exceptuando las zonas polares (Givnish et al., 2015; Kim et al., 2020; Pérez-Escobar et

al., 2024; Schmid & Dressler, 1981). Estas plantas presentan diversos hábitos de crecimiento, pudiendo ser terrestres, rupícolas, epífitas e incluso trepadoras. Además, las orquídeas son conocidas por tener una gran diversidad de estrategias para la polinización, que se pueden resumir en dos grandes grupos: La polinización por engaño, ya sea sexual o alimenticio, y la polinización por recompensa (Karremans, 2023). La capacidad de utilizar diferentes sustratos y tener gran diversidad de estrategias para la polinización, ha facilitado un proceso acelerado de diversificación en múltiples linajes, especialmente en las regiones tropicales, donde Centroamérica representan uno de los hotspots de diversidad de orquídeas a nivel mundial (Bogarín et al., 2013; Pérez Escobar et al., 2017; Pérez-Escobar et al., 2024).

La diversidad de orquídeas neotropicales se estima en cerca de 13,000 especies (Pérez Escobar et al. 2024). Al menos el 10% de estas especies es polinizado por un grupo especializado de abejas conocidas como "abejas de las orquídeas" (Apidae: Euglossini) (Ramírez et al., 2002; Roubik & Hanson, 2004). Estas orquídeas emplean principalmente dos mecanismos de atracción: el engaño alimenticio y la recompensa (compuestos aromáticos). En la polinización por engaño, las orquídeas han desarrollado adaptaciones engañosas de tipo morfológicas, químicas y/o sensoriales que permiten estimular el sistema nervioso de los polinizadores y, en el momento de la visita floral, puede haber remoción o deposición del polen, pero el polinizador no obtiene ninguna recompensa (van der Pijl & Dodson, 1966; Dressler, 1981; Ackerman, 1986, Karremans, 2023). En el caso de la polinización por engaño alimenticio, las orquídeas se asemejan a plantas que ofrecen algún recurso para las abejas, como aceites, polen o néctar, pero al ingresar a las flores, no hay recursos disponibles (Karremans, 2023). Mientras tanto, en la polinización por recompensa, las orquídeas están ofreciendo algún tipo de recursos para la alimentación de los polinizadores, aprovisionamiento de los nidos o reproducción. Entre las recompensas más comunes está el polen, néctar, aceites y compuestos aromáticos. Este último recurso ha sido una ventaja adaptativa clave, permitiendo una rápida diversificación de las orquídeas mediante la especialización en la polinización exclusiva por abejas euglossinas (Givnish et al., 2015; Pérez Escobar et al., 2017; Ramírez et al., 2011).

Generalidades de las abejas Euglossinas y su relación con las orquídeas:

Las abejas de las orquídeas forman parte del clado de las abejas corbiculadas junto con Apini, Bombini y Meliponini (Michener, 2007; Roubik & Hanson, 2004). Estas abejas están distribuidas de manera natural en la región neotropical, abarcando desde el norte de México hasta el norte de Argentina y Chile, aunque algunas especies han sido

introducidas y se han adaptado al sur de los Estados Unidos, principalmente en Florida y Texas (Gonzalez et al., 2017; Griswold et al., 2015; Minckley & Reyes, 1996; Skov & Wiley, 2005). Actualmente, se reconocen al menos 200 especies agrupadas en cinco géneros: *Aglæ*, *Eufriesea*, *Eulaema*, *Euglossa* y *Exaerete*. Las abejas euglossinas son componentes clave de los bosques neotropicales, poseen amplios rangos de vuelo, cubriendo hasta 20 km por día (Janzen, 1971), lo que les permite visitar y polinizar una gran diversidad de plantas, incluidas al menos 40 familias (Ramírez et al., 2002), y alrededor de 1,300 especies de orquídeas (Ramírez et al., 2002; Roubik & Hanson, 2004).

Colombia alberga la mayor diversidad de abejas euglossinas con 113 especies registradas, mientras que Costa Rica es el segundo país más diverso de Centroamérica con 66 especies, detrás de Panamá, que cuenta con 69 (Ramírez et al., 2002). Las abejas de las orquídeas se distribuyen en ambientes cálidos y húmedos, desde el nivel del mar hasta los 2000 m.s.n.m. (Gonzalez & Engel, 2004; McCravy et al., 2016; Vélez & Pulido, 2005). Sin embargo, en los últimos años se han reportado especies por encima de estos límites altitudinales (Nates et al., 2006). La mayor riqueza de especies de Euglossini se encuentra en bosques bien conservados de elevaciones intermedias (Cameron, 2004), donde representan aproximadamente el 25% de las especies de abejas (Dick et al., 2004). Estas abejas también son utilizadas como bioindicadores de la calidad ambiental y pueden habitar en ecosistemas agrícolas y urbanos, siempre que la disponibilidad y calidad de los recursos alimenticios y sitios de nidificación sean adecuados (Añino et al., 2019; da Rocha-Filho et al., 2020; Hedström et al., 2006). En Costa Rica, se encuentran representantes de todos los géneros de abejas euglossinas, excepto el género monotípico *Aglæ* (McDonald et al., 2022).

Los machos de las abejas euglossinas desempeñan un papel crucial en la polinización de varios géneros de orquídeas, siendo los principales polinizadores de numerosas especies pertenecientes a las subfamilias Epidendroideae y Vanilloideae. Entre los géneros polinizados por estas abejas se encuentran *Catasetum*, *Cycnoches* y *Mormodes* en la subtribu Catasetinae; *Acineta*, *Coryanthes*, *Gongora* y *Stanhopea* en la subtribu Stanhopeinae; *Macroclinium* y *Notylia* en la subtribu Oncidiinae; *Cochleanthes*, *Dichaea* y *Keferstenia* en la subtribu Zygopetalinae; así como *Vanilla* en la subtribu Vanilleae, entre otros géneros (Ackerman et al., 2023). Los machos son atraídos por las sustancias aromáticas producidas por estas orquídeas y, al ingresar a las flores, los polinarios (estructuras que contienen todo el polen de una flor) se adhieren a diferentes partes de sus cuerpos (Roubik & Hanson, 2004). Posteriormente, cuando

los machos visitan otra flor de la misma especie, depositan el polinario, permitiendo así la polinización y reproducción de la orquídea.

La diversificación de las orquídeas y las abejas euglossinas ocurrió de manera paralela, hace aproximadamente 25 y 18-20 millones de años respectivamente, con interacciones que pueden ser altamente específicas (Cameron, 2004; Kim et al., 2020; Roubik & Ackerman, 1987; Schmid & Dressler, 1981). Las abejas euglossinas son ecológicamente importantes y han sido objeto de numerosos estudios debido al comportamiento de los machos de recolección de compuestos aromáticos, utilizados en su reproducción, donde los perfumes funcionan como señal sexual para estimular a las hembras (Henske et al., 2023). Este comportamiento facilita su estudio, lo que ha permitido obtener información sobre sus interacciones con orquídeas, diversidad y dinámicas poblacionales (Margatto et al., 2019; Roubik & Ackerman, 1987; Singer et al., 2008; Zimmermann et al., 2011).

Fenología de las abejas Euglossinas y su interacción con las orquídeas:

La fenología es el estudio de los procesos biológicos recurrentes en los seres vivos y su relación con factores ambientales como la precipitación, la temperatura y los cambios que ocurren en las poblaciones a lo largo de un año o más (Kaesha & Jianguo, 2006). Los estudios fenológicos son cruciales para comprender la disponibilidad de recursos para las abejas en diferentes épocas del año y cómo los factores ambientales influyen en su diversidad y abundancia (Chapurlat et al., 2020; Sabat & Ackerman, 1996; Sletvold et al., 2010). Por ejemplo, se ha demostrado que un aumento en la disponibilidad de recursos florales incrementa la diversidad de abejas (Groom & Schwarz, 2011; Klein et al., 2003), aunque la floración de muchas plantas está condicionada por factores climáticos como la temperatura y la precipitación.

En las zonas templadas se ha investigado ampliamente la fenología de las abejas solitarias, especialmente en especies oligoléticas que emergen durante la floración de las plantas que polinizan (Danforth et al., 2019). Sin embargo, en las zonas tropicales la fenología de la mayoría de las abejas solitarias sigue siendo poco conocida. Un ejemplo relevante son las especies estacionales y univoltinas de *Eufriesea*, que emergen al inicio de la temporada de lluvias tras pasar la mayor parte del año en diapausa (Margatto et al., 2019). Otras especies de abejas euglossinas, al ser multivoltinas, están presentes durante todo el año, con su abundancia vinculada a los patrones de precipitación y la disponibilidad de recursos florales (Margatto et al., 2019; Rocha & Garófalo, 2014). Estudios recientes han analizado la fenología de las abejas euglossinas y sus variaciones a lo largo del año, encontrando que varias especies

muestran picos de abundancia en diferentes épocas. Las diferencias estacionales se atribuyen principalmente a la variación en las coberturas vegetales y a los niveles de precipitación. (Lopes et al., 2022; Pérez-Buitrago et al., 2022; Rasmussen, 2009).

A pesar de estos avances, solo se ha realizado un estudio en Panamá sobre la relación entre las abejas euglossinas y las orquídeas, mediante el análisis de los polinarios adheridos a las abejas (Roubik & Ackerman, 1987). Este estudio no abordó la dinámica temporal de la interacción planta-abeja, a pesar de contar con un esfuerzo de muestreo significativo. En general, se sabe poco sobre cómo la variabilidad climática afecta la dinámica de la interacción abeja-orquídea a lo largo del año.

Desarrollar un estudio sistemático sobre esta interacción permitirá comprender mejor las relaciones entre especies, generar nueva información sobre la fenología de las orquídeas y su interacción con las abejas, y analizar la variación funcional de la comunidad de abejas a lo largo del año. La hipótesis de este trabajo plantea que la precipitación es el factor ambiental clave que influye en la fenología reproductiva de las orquídeas y en la composición de la comunidad de abejas euglossinas, determinando así la dinámica de interacción reproductiva entre ambos grupos a lo largo del año.

Para comprender mejor la dinámica de las interacciones entre las abejas euglossinas y las orquídeas, en relación con la variación climática, planteamos los siguientes objetivos:

Objetivo general.

Caracterizar la dinámica de la comunidad de abejas euglossinas y la fenología reproductiva de las orquídeas que estas polinizan en un bosque húmedo tropical de tierras bajas del caribe de Costa Rica.

Objetivos específicos.

1. Comparar la riqueza de especies de abejas euglossinas y su abundancia relativa a lo largo del año y su variación respecto a la precipitación.
2. Determinar la fenología reproductiva anual de las especies de orquídeas polinizadas por abejas euglossinas con base en la presencia de polinarios adheridos a las abejas.
3. Construir una red bipartita de interacciones mutualistas entre las abejas euglossinas y las orquídeas que polinizan.

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SEASONALITY AND RELATIVE ABUNDANCE OF ORCHID BEES (APIDAE: EUGLOSSINI) IN A TROPICAL RAIN FOREST IN COSTA RICA

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ABSTRACT:

Euglossini bees pollinate many species of orchids and other plants in the tropical forest. Studies show that there is no seasonality for many Euglossini species except for *Eufriesea*, which is influenced by rainfall patterns. The research was carried out at the La Selva Biological Station, Costa Rica, which is characterized by a humid tropical climate. Male Euglossini bees were sampled monthly in the tropical rain forest at the La Selva Biological Station in Costa Rica to test for seasonality and relative abundance of orchid bees. In a period of 12 months, a total of 4,994 male Euglossini bees were captured using chemical baits, distributed in 4 genera and 31 species. *Euglossa purpurea* was the most abundant species. The dry season, from the month of January to April, presented less diversity and abundance than did the rainy season, from the month of May to December. Seasonality patterns were found in 20 species, and 4 showed significant correlations with precipitation. A statistical model confirmed that precipitation significantly influences the abundance of bees. Significant differences were also detected between climatic seasons for some bee species. Costa Rica has approximately 65 species of Euglossini bees, 31 of which have been recorded at the La Selva Biological Station, including new records for *Eufriesea* and *Euglossa*. The communities of these bees vary dynamically and are influenced by precipitation, affecting their abundance and patterns of seasonality. Studies suggest that climate

change may influence these dynamics, highlighting the need for long-term research to conserve these pollinators and the plants that depend on them.

INTRODUCTION:

Phenology involves all the cyclic processes that occur during the life of an organism and its relationship with environmental variables (Kreitzman and Foster, 2010; Visser et al. 2010). Phenology, together with other factors, determines the presence of individuals in ecosystems (Hodgson et al., 2011). In general, the relative abundance of insects' changes throughout the year in response to climatic transitions that occur over time (Memmott et al., 2007; Visser et al., 2010). Many organisms have periods of greater or lesser activity, and all these processes are influenced by the natural history of populations, phylogeny, availability of resources and climate (Cortés-Flores et al., 2017).

Euglossine bees or orchid bees are a group of corbiculate bees that are naturally distributed from northern Mexico to northern Argentina with approximately 250 species (McCravy et al., 2016; McDonald et al., 2022). These bees are important as they pollinate at least 10% of all neotropical orchids as well as many other angiosperms (Cameron, 2004; Givnish et al., 2015). Their wide flight range (~20 km / day) promotes gene flow between different plant populations, making them valuable components of tropical forests (Añino et al., 2019; Margatto et al., 2019). Euglossine bees have been extensively studied because males collect fragrances for use during the process of courtship with females (Dressler, 1982; Henske et al., 2023; Ramírez et al., 2002).

Bee phenology is important because of its dependence on floral resources and because bees are the main pollinators in many ecosystems (Pinto et al., 2019; Torezan-Silingardi et al., 2021). Most research on the phenology of bee species has been carried out in temperate seasonal ecosystems, where bees exhibit an almost synchronous appearance with the flowering periods of diverse plant species (Bertrand et al., 2019; Neave et al., 2020; Ogilvie et al., 2017). In addition, the bee species studied are mostly oligolectic and depend on a very select group of plants with short flowering periods, during which time they feed and provision their nests (Minckley, 2008; Smith et al., 2019).

In the Neotropics, seasonal abundances of bees have been conducted mostly in dry forest ecosystems, where the periods of rainfall are very well defined (Hedström et al. 2006; Margatto et al. 2019; Flórez-Gómez et al. 2020; Darragh et al. 2023). However, studies designed to study the phenology of the community of orchid bees are still scarce. The studied species of *Eufriesea* are seasonal, the adults being present at the beginning of the rainy season (Dressler, 1982; Roubik & Hanson, 2004). For the remaining genera various factors may influence their abundance, for example reproduction periods

(Vilhena et al., 2017), greater availability of resources (Rocha & Garófalo, 2014) or other environmental conditions (Bravo et al., 2022; Ramírez et al., 2015; Roubik & Ackerman, 1987). Previous works have shown that precipitation is an important factor affecting the abundance and diversity of Euglossini bees (Dressler, 1982; Roubik & Hanson, 2004). Climatic seasonality can determine the periods with the greatest and least abundance of different bee species and, in turn, the availability of the pollination services they provide. However, the question remains how the phenology of bee communities varies in environments with low seasonality. Many tropical forests have very constant temperatures throughout the year and little variation in precipitation. This work aims to characterize the dynamics of the euglossine bee community and the phenology of these bees in a tropical humid forest of Costa Rica.

We hypothesize that precipitation is the environmental factor that most affects the presence of Euglossini bees in a tropical rainforest. As rainfall increases, a decrease in the abundance and diversity of Euglossini bees will be observed. This is because heavy rainfall could limit the bee's foraging activity and affect the availability of floral resources. Consequently, the abundance of orchid bees is expected to be higher in the dry season compared to the rainy season. If precipitation is indeed the most influential environmental factor, the abundance and diversity patterns of Euglossini bees should vary significantly between the two seasons, demonstrating a direct dependence on this factor.

MATERIALS AND METHODS:

Study site:

The study was carried out at the La Selva Biological Station (LSBS), which is located on the slopes of the Braulio Carrillo National Park in the central volcanic mountain range of Costa Rica. It is owned by the Organization for Tropical Studies (OTS) and is in a tropical humid forest ecosystem (Holdridge, 1967) located in Puerto Viejo de Sarapiquí, Heredia, Costa Rica (**Figure 1**). It has an average elevation of 35 m above sea level and an area of 1,536 hectares. It shows a slightly bimodal pattern of rainfall of approximately 4,000 mm/year, where July and December are the months with the highest rainfall, > 500 and > 400 mm, respectively. The driest months are March and September, < 200 and < 350 mm, respectively (Sanford et al., 1994). The temperature throughout the year is constant (~ 25 °C).

Bee sampling was carried out from February 2023 to January 2024. Each month, we sampled three points in mature forest (at least 40-50 years of protection) during three

consecutive days. Sampling points were separated by approximately 3 km from each other and remained constant between samplings.

Euglossini male sampling and climatic data:

The sampling of the bees was carried out from 8 am to 12 md, the time of greatest activity of the Euglossine bees. Six aromatic compounds, that are classified like “good attractants” were used as baits, three liquid solutions (eugenol, methyl salicylate and eucalyptol), and three in crystalline state (vanillin, skatole and 1,4 dimethoxybenzene). Liquid baits were placed in cotton balls hanging from a synthetic rope at a height of ~1.5 m. Crystalline baits were macerated and placed at ground level over rocks, concrete or fallen wood pieces. Every half hour all males present at the baits were captured, placed in 50 ml centrifugation tubes in a cooler with ice for a few minutes to reduce metabolism and facilitate the manipulation of the bees. Sampling was nonlethal as all bees were classified to species directly in the field (after previous training to do so) and then released. For newly collected species, morphospecies were assigned and reference specimens were collected for later identification. Voucher specimens were deposited in the Museum of Zoology, Biodiversity and Tropical Ecology Research Center at the University of Costa Rica. The identification was carried out using Roubik & Hanson (2004) and comparisons with museum specimens. Daily abundance of each species was registered. To avoid recounting, bees were marked with model paint (Testor®) on the dorsal part of the mesosoma. This was done to avoid recounts.

The climate data were obtained from the automatic climate stations of the LSBS through the platform owned by the Organization for Tropical Studies. The data used for the analyses were the accumulated averages of the last 53 years (**Figure 2**). For the present work, two seasons were defined: low rainy season (LRS), which begins in January and ends in April, with a rainfall of < 300 mm per month, and high rainy season (HRS), which begins in May and ends in December, with rainfall > 300 mm per month.

Statistical analysis:

For all analyses and graphs, R software (R Core Team, 2024) was used. Each of the packages used is mentioned directly in each analysis.

To determine the completeness of our sampling rarefaction curves were generated with the first three orders of Hill numbers (q) (Hill, 1973), where the sensitivity to the abundance of the species increases as the parameter q increases. Therefore, $q = 0$ indicates the accumulated richness of species, and the abundance is independent for

this analysis. At $q = 1$, the effective number of common species is shown, where the importance of each species is proportional to the abundance. At $q = 2$, the effective number of dominant species is shown, where those that are more dominant have greater values than rare species (Chao et al., 2014). We also compared species richness between the dry and rainy seasons at LSBS. Analyses were done using the iNEXT package (Hsieh et al., 2016).

To determine the seasonality of each of the species, we used the monthly relative abundances of each species and calculated circular statistics analyses. Months were used as angles, with an interval of 30° (12 months), with February 2023 as 30° and January 2024 as 360° . To test seasonality in the peak abundance of each species, we used the statistics proposed in Morellato et al. (2000). For this, the abundance date was converted to an angle, and then the mean angle and angular standard deviation were calculated. The Rayleigh test was subsequently applied to determine if the abundance is uniformly distributed throughout the year. A value close to 0 indicates that the data are uniformly distributed throughout the year, whereas a value close to 1 indicates that the data are concentrated at a particular moment of the year (Martins et al., 2021; Morellato et al., 2000).

Linear regressions were performed to determine the effects of precipitation on the abundance for each subgenus (*Euglossa* and *Glossura*) and species. Student's t test was carried out to evaluate whether the monthly abundance averages of each species varied significantly between seasons. A mixed generalized linear model with a zero-inflated Poisson distribution was constructed, where precipitation and species were randomly included, to test the hypothesis that precipitation has a significant effect on the abundance of bees. This model was compared with the intercept of a null model via the Akaike information criteria (AIC), and if the model with the observed data had a better fit than the null model, the model offered a better description of the data.

RESULTS:

We sampled 4,994 males belonging to four genera and 31 species of Euglossini. The most abundant genus was *Euglossa* Friese 1899, with 21 species, and the most dominant species throughout the year was *Euglossa purpurea*, with 1,762 individuals. The month with the greatest diversity of species was April (26 species), while June had the highest abundance (674 individuals). On the other hand, February, October and November were the months with the least number of species, with 16 species each (**Table 1**). The asymptote in the rarefaction curves was reached for the whole year

(**Figure 3A**) and for each season (**Figure 3B**). The LRS showed a greater abundance, dominance and species richness (30 species) than the HRS, in which 26 species were recorded (**Figure 3B**). In the following we will use contractions of the generic names: *Eufriesea*: *Ef.*, *Euglossa*: *Eg.*, *Eulaema*: *El.*, *Exaerete*: *Ex.* Species absent in the rainy season were *Ef. anisochlora*, *Ef. schmidtiana*, *El. polychroma* and *Eg. townsendi*. The rest of the species were present in both seasons.

All species collected in this study showed different relative abundances and abundance peaks in each month. These data were included in the analyses to determine the seasonality of each species. We found patterns for seasonality for 18 species ($p < 0.001$ ***) (**Figure 4**). Compared with the other species, *Ef. ornata* and *Ef. rufocauda* presented the highest mean values of the vector (0.93 and 0.89, respectively), which indicates that the data were concentrated in a particular period of the year (June and May, respectively) (**Table 2**).

Of all the species, only four had an abundance significantly correlated with precipitation: *Eg. allosticta* ($r^2 = 0.255$, $p < 0.01$), *Eg. dodsoni* ($r^2 = 0.392$, $p < 0.05$), *Eg. mixta* ($r^2 = 0.362$, $p < 0.05$), and *Eg. dilemma* ($r^2 = 0.542$, $p < 0.01$) (**Figure 5**). Although we found highly significant differences in precipitation between the LRS and HRS in La Selva ($t = 33.99$, $df = 284.16$, $p < 0.001$), no significant differences were observed with respect to temperature ($t = 2.79$, $df = 5.69$, $p > 0.08$); therefore, this variable was not included in the additional statistical analyses (**Figure 6**).

Student's *t* tests revealed significant differences in the abundance between seasons for only four species: *Eg. purpurea* ($t = -3.47$, $df = 9.26$, $p < 0.05$), *Eg. allosticta* ($t = 4.92$, $df = 10.77$, $p < 0.001$), *Eg. mixta* ($t = -3.92$, $df = 10.81$, $p < 0.05$), and *Eg. championi* ($t = -3.66$, $df = 10.74$, $p < 0.05$) (**Figure 7**). The mixed generalized linear model revealed that precipitation had a highly significant effect on the abundance of Euglossini bees ($p < 0.001$ ***). A summary of the model is presented in **Table 1**.

DISCUSSION:

Diversity patterns:

Costa Rica is one of the countries with highest diversity of orchid bees, having approximately 65 species (Ramírez et al., 2002; Roubik & Hanson, 2004). For LSBS, 31 species were found, including nine newly recorded species. In the present study, we did not find the species *Ef. macroglossa*, *El. bombiformis*, *Eg. cognata*, *Eg. crassipunctata* and *Eg. flammea*, which were previously reported to occur in the station in other works

(Brenes-D, 2023 personal communication). In all, a total of 36 Euglossini species have been recorded to occur in LSBS (55.4% of the species known to occur in the country). Localities with the most diverse communities of euglossine bees are found in Panama (Roubik & Ackerman, 1987) and in Colombia (Ramírez et al., 2015), each with approximately 50 species in each site. Euglossine bee communities are dynamic throughout their distribution and studies have reported the increase or decrease in the numbers of species (Ackerman, 1983; Bravo et al., 2022; Margatto et al., 2019; Nemésio & Silveira, 2007), and even pronounced sudden declines in the number of species (Vega-Hidalgo et al., 2020).

Effect of precipitation on the abundance of euglossine bees:

The results of the current study agree with results of prior investigations that related the abundance of Euglossini bees to periods of precipitation. Ackerman (1983) reported that the only species that was more abundant in the drier months in Panama was *Eg. dodsoni*. In our study, we found that that only *Eg. cybelia* and *El. cingulata* were more abundant in the LRS. In contrast, most of the species we found in this study were more abundant in the HRS. In Brazil, Nemésio & Vasconcelos (2013) found that euglossine bees tend to be more abundant and active in the dry season, whereas they reduce their foraging activity in the rainy season, as has been reported for other insect pollinators in the Neotropics (Bascompte & Jordano 2007a). Recent research suggests that seasonal fluctuations, such as in temperature and precipitation, can influence the periods of activity of these bees, especially in regions such as the Brazilian Amazon where much of the flowering occurs during the wet season (Sofia et al., 2004).

Changes in the monthly abundances of some of the species in this study have already been reported in other studies, where most of the species presented abundance peaks at the end of the dry season and the beginning of the rainy season (Pearson & Dressler, 1985; RochaFilho & Garófalo, 2014). In Panama, species such as *Eg. allosticta*, *Eg. azueroviridis*, *Eg. cognata*, *Eg. crassipunctata*, *Eg. cybelia*, *Eg. despecta*, *Eg. dissimula*, *Eg. dressleri*, *Eg. hansonii*, *Eg. heterosticta*, *Eg. imperialis*, *Eg. mixta*, *Eg. sapphirina*, *Eg. tridentata*, *Eg. variabilis*, *Ef. corusca*, *Ef. lucifera*, *Ef. pulchra*, *Ef. surinamensis*, *El. cingulata*, *El. meriana*, *El. nigrita* and *Ex. smaragdina* showed abundance peaks during the transition from the dry to the rainy season (Ackerman, 1983), as did the species reported by Ramírez (2019). In our study, we also found that *Eg. asarophora*, *Eg. bursigera*, *Eg. dilemma*, and *Ex. frontalis* showed abundance peaks during the transition from the LRS to the HRS. Other species such as *Ef. anisochlora*, *Ef. ornata*, *Ef. schmidtiana* presented abundance peaks during the rainy season, such as *Eg. decepatrix*,

Eg. dodsoni, *Eg. hemichlora*, *Eg. sapphirine* in our study (Ackerman, 1983). The rest of the species did not have a marked abundance peak, but were mainly present in the HRS. Variations in the abundance of Euglossini bees can be influenced by different factors, such as nest pathogens or natural enemies (Dressler, 1982), the level of environmental disturbance (Añino et al., 2019; Brosi, 2009; Pinto et al., 2019), or the availability of floral resources (Rocha & Garófalo, 2014).

Effect of precipitation on the seasonality of euglossine species.

The species of *Eufriesea* had a seasonal pattern, with abundance peaks at the end of the dry season and the beginning of the wet season, as has been reported in Brazil (Vilhena et al. 2017, Margatto et al. 2019), Colombia (Ramírez et al. 2015), and Honduras (McCravy et al., 2016). However, species of *Eufriesea* are not the only euglossines to show patterns of seasonality. The abundances of *Eg. cordata*, *Eg. iopoecila*, *Eg. sapphirina*, *Eg. stellfeldi*, *Eg. truncata* and *El. nigrita* were concentrated at specific times of the year in Piciguaba, Brazil, which was also the case for *Ef. violacea*, *Eg. cordata*, *Eg. fimbriata* and *El. nigrita* in Palotina, Brazil (Rocha & Garófalo, 2014).

This is the first work that reports seasonal patterns in four genera and more than 7 species. *Euglossa* presented different patterns of seasonality; some species, such as *Eg. allosticta*, *Eg. dodsoni*, *Eg. mixta* and *Eg. purpurea*, were much more abundant and dominant during the high-rainfall season. However, Ackerman (1983), reported that *Eg. allosticta* was more abundant in the transition from the dry to the rainy season; likewise, *Eg. purpurea* was more abundant in the dry season, whereas in our study, it was dominant during the HRS. Species such as *Eg. championi* and *Eg. variabilis* were more abundant in the LRS at LSBS, but they were more abundant in the dry season in Panama. *Euglossa ignita* was present mainly in the dry season in Brazil (Brito & Rêgo, 2001), whereas this species was present mainly in the rainy season in the current study. For species of *Eulaema* and *Exaerete* the relative abundances throughout the year were very stable, but *El. cingulata* was present in the period with less humidity in Brazil (Rocha & Garófalo 2014). Apparently, there is a pattern where the phenology of some euglossine bees varies according to the changes in precipitation, but most of the species do not show clear patterns of seasonal abundance. Long-term studies with Euglossini in Panama have shown that the species can have different seasonal patterns over the years, but the species are always present in traps (Roubik & Ackerman 1987).

Rare and less common species may be more susceptible to changes in periods of precipitation because they have higher energy requirements, more specific ecological niches or flower resources that are only available during periods of drought (da Rocha-

Filho et al., 2020; LeãoGomes & Vasconcelos, 2023; Miller-Struttmann, 2024). Thus, it is possible that these species are not present in the ecosystems studied or that the fragrances used in this work do not attract them.

The results of the linear regressions indicate that the abundance of 13% of the bee species studied are significantly positively correlated with precipitation. These findings coincide with previous studies that have documented similar effects of precipitation on the abundance of euglossine bees (Enríquez-Espinosa et al., 2022; Rocha-Filho & Garofalo, 2013). In addition, euglossine species can show different seasonal preferences, with some being abundant during dry periods, whereas others reach their highest abundance during wet periods (Margatto et al., 2019). This explains the existence of population dynamics, in addition to the fact that some species may become more sensitive to changes in forest composition and cover, disturbances and the effects of climate change (da Silva Carneiro et al., 2022). Reproductive cycles could be a key factor in the abundance of euglossine bees and their variation throughout the year. However, this factor has not been studied and we suggest that it be considered in future studies. These results highlight precipitation as a key factor in the phenology of euglossine bees, with important implications for the ecology of pollination and plant–pollinator interactions in tropical ecosystems.

In conclusion, our results highlight that precipitation is a critical factor influencing the abundance and phenology of euglossine bees in tropical ecosystems. We observed that several species showed a positive correlation between abundance and rainfall levels, in line with findings from previous studies. This relationship suggests that rainfall patterns not only influence the seasonal activity of these bees, but also the dynamics of plant–pollinator interactions, as some species are most abundant during wet periods, while others are more active during dry periods. These findings have important ecological implications, as changes in the rainfall regime, possibly due to climate change, could alter the distribution and abundance of these bees, affecting the stability of pollinator communities in the tropics and their interactions with plants.

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SUPPLEMENTARY MATERIAL:

Table 1. Total samples collected in La Selva Biological Station during 2023-2024
(Species richness/Abundance)

Month	<i>Eufriesea</i>	<i>Euglossa</i>	<i>Eulaema</i>	<i>Exaerete</i>
January	0/0	16/295	3/36	1/3
February	0/0	16/174	1/3	2/6
March	1/2	17/175	2/25	2/9
April	3/25	19/387	2/30	2/22
May	1/63	15/459	2/26	2/13
June	2/30	18/614	2/13	2/17
July	2/12	17/408	2/13	2/9
August	2/10	17/433	2/17	2/7
September	0/0	16/408	2/19	2/6
October	0/0	14/205	2/5	2/6
November	0/0	13/352	2/30	2/8
December	0/0	17/380	2/15	2/3

Table 2. Average vector length and Rayleigh test results for each species. Bold numbers represent those species that had significant test results.

<i>Euglossine species</i>	<i>Average vector length</i>	<i>Rayleigh test</i>
<i>Euglossa purpurea</i>	0.299	<0,001
<i>Euglossa imperialis</i>	0.05	0,165
<i>Euglossa dodsoni</i>	0.20	<0,001
<i>Euglossa allosticta</i>	0.46	<0,001
<i>Euglossa sapphirina</i>	0.20	<0,001
<i>Eulaema meriana</i>	0.06	0,43
<i>Euglossa mixta</i>	0.31	<0,001

<i>Euglossa ignita</i>	0.24	<0,001
<i>Euglossa ignita</i>	0.65	<0,001
<i>Euglossa gorgonensis</i>	0.49	<0,001
<i>Euglossa tridentata</i>	0.89	<0,001
<i>Eufriesea rufocauda</i>	0.14	0,11
<i>Euglossa championi</i>	0.23	<0,05
<i>Eulaema cingulata</i>	0.19	0,057
<i>Euglossa villosiventris</i>	0.48	<0,001
<i>Euglossa cybelia</i>	0.29	0,002
<i>Exaerete frontalis</i>	0.40	<0,001
<i>Euglossa bursigera</i>	0.22	0,079
<i>Euglossa variabilis</i>	0.049	0,9
<i>Euglossa villosa</i>	0.30	<0,05
<i>Euglossa asarophora</i>	0.46	<0,001
<i>Exaerete smaragdina</i>	0.54	<0,001
<i>Euglossa hemichlora</i>	0.42	<0,05
<i>Euglossa heterosticta</i>	0.68	<0,001
<i>Euglossa viridisima</i>	0.93	<0,001
<i>Eufriesea ornata</i>	0.93	<0,001
<i>Euglossa decepatrix</i>	0.6	<0,05
<i>Eufriesea anisochlora</i>	1	0.13

<i>Eufriesea schmidtiana</i>	1	0.51
<i>Eulaema polychroma</i>	1	0.51
<i>Euglossa townsendi</i>	1	0.51

Tabla 3. Summary of the generalized linear mixed model with Poisson distribution of inflated zeros.

STIMATES STANDAR Z P

ERROR

SPECIES				
INTERCEPT	1.333	0.219	5.921	<0.001***
RAINFALL	0.002	0.00013	13.74	<0.001***

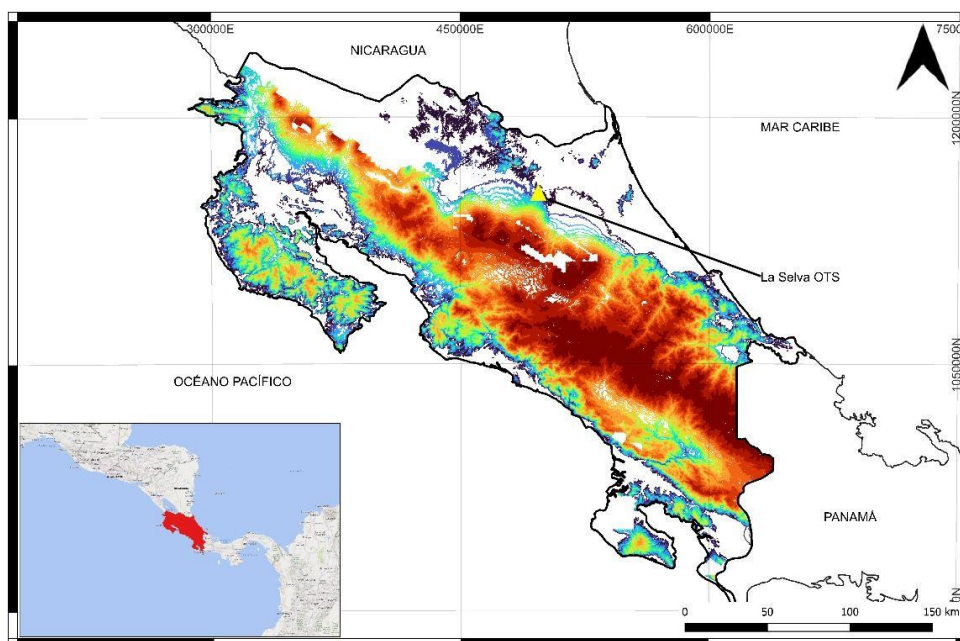


Figure 1. Geographical location of La Selva Biological Station, owned by the Organization for Tropical Studies (OTS).

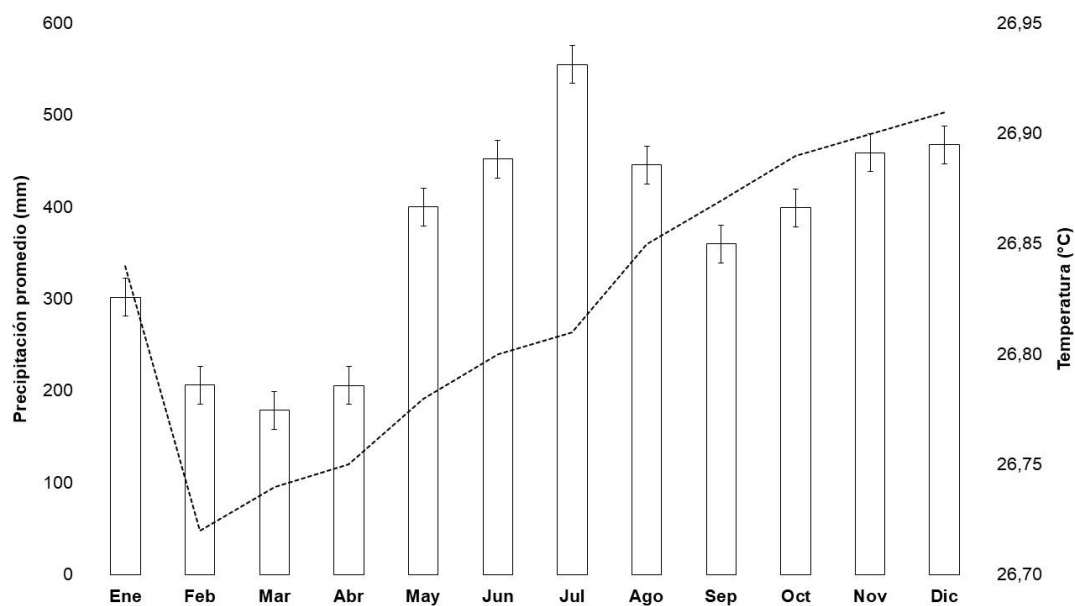


Figure 2. Average monthly precipitation at the La Selva Biological Station. Data collected from 1963 to 2016. Bars represent average monthly precipitation (mm). The dotted line represents the average monthly temperature (°C).

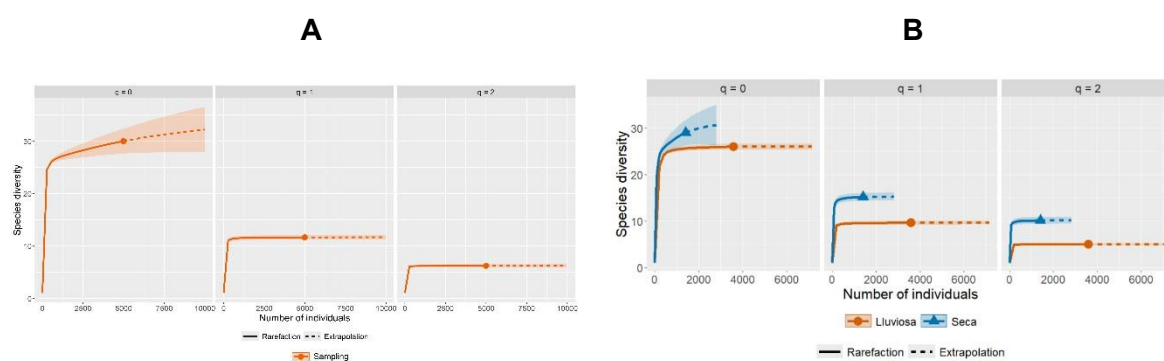
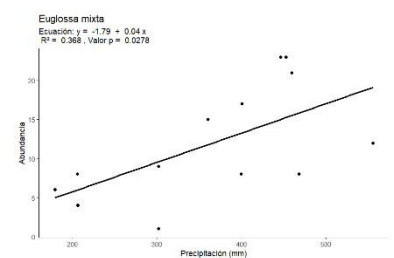
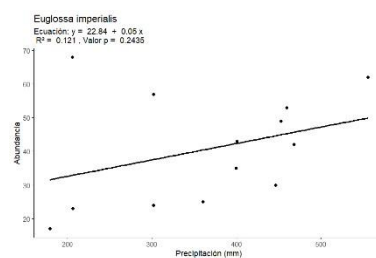
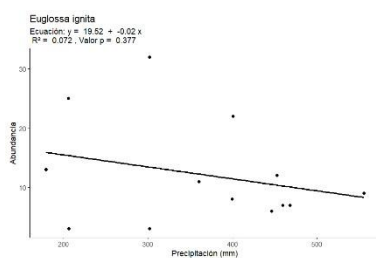
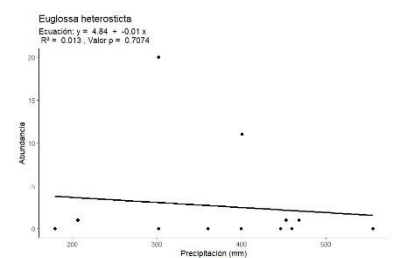
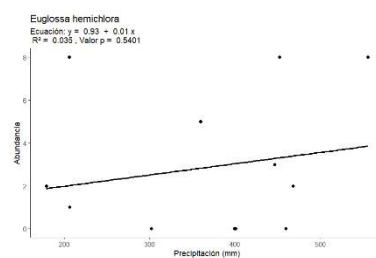
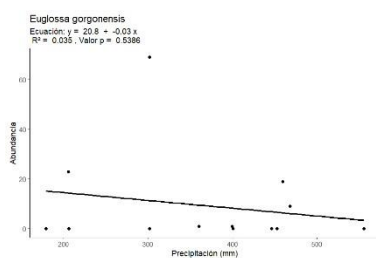
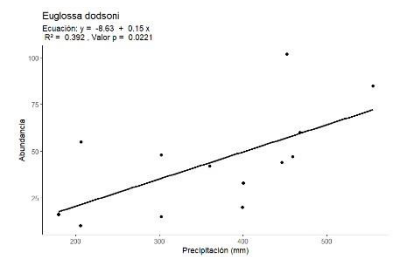
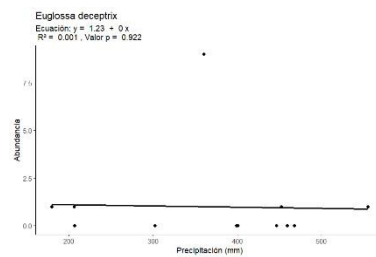
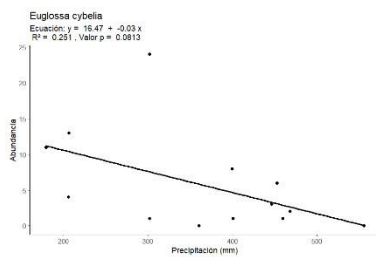
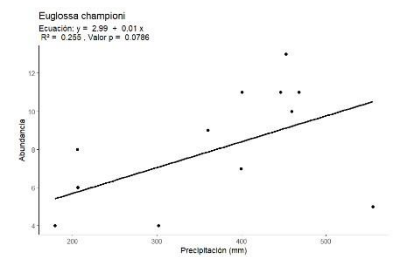
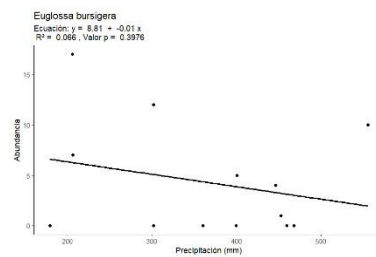
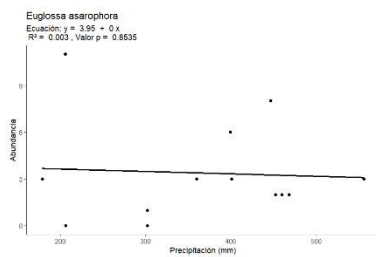
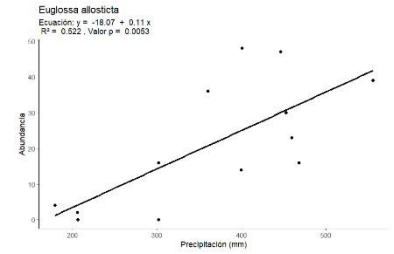
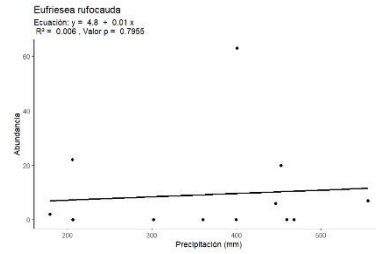
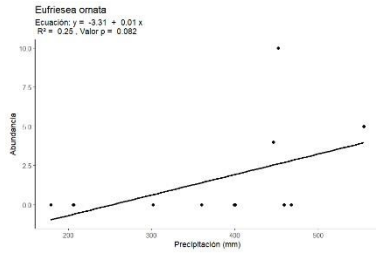


Figure 3: A. Completeness of sampling for La Selva Biological Station during the sampling period 2023–2024. **B.** Bee species diversity in HRS and LRS at the La Selva Biological Station, Costa Rica. Graphs show (A) species richness (Hill number for $q = 0$), (B) equally abundant species ($q = 1$), (C) dominance ($q = 2$) and (D) sample coverage curve. Diversity curves were constructed using species abundances (solid lines) and extrapolated abundances (dashed lines) with estimates based on sample size. Colored shaded regions represent 95% confidence intervals.

cingulata Q) *Ex. fontalis*. R) *Ex. smaragdina* S) *Eg. championi*. T) *Eg. heterosticta*. U)
Eg. imperialis. V) *Eg.*
purpurea. W) *Eg. variabilis*. X) *Eg. villosa*. Y) *El. cingulata*. Z) *El. meriana*.



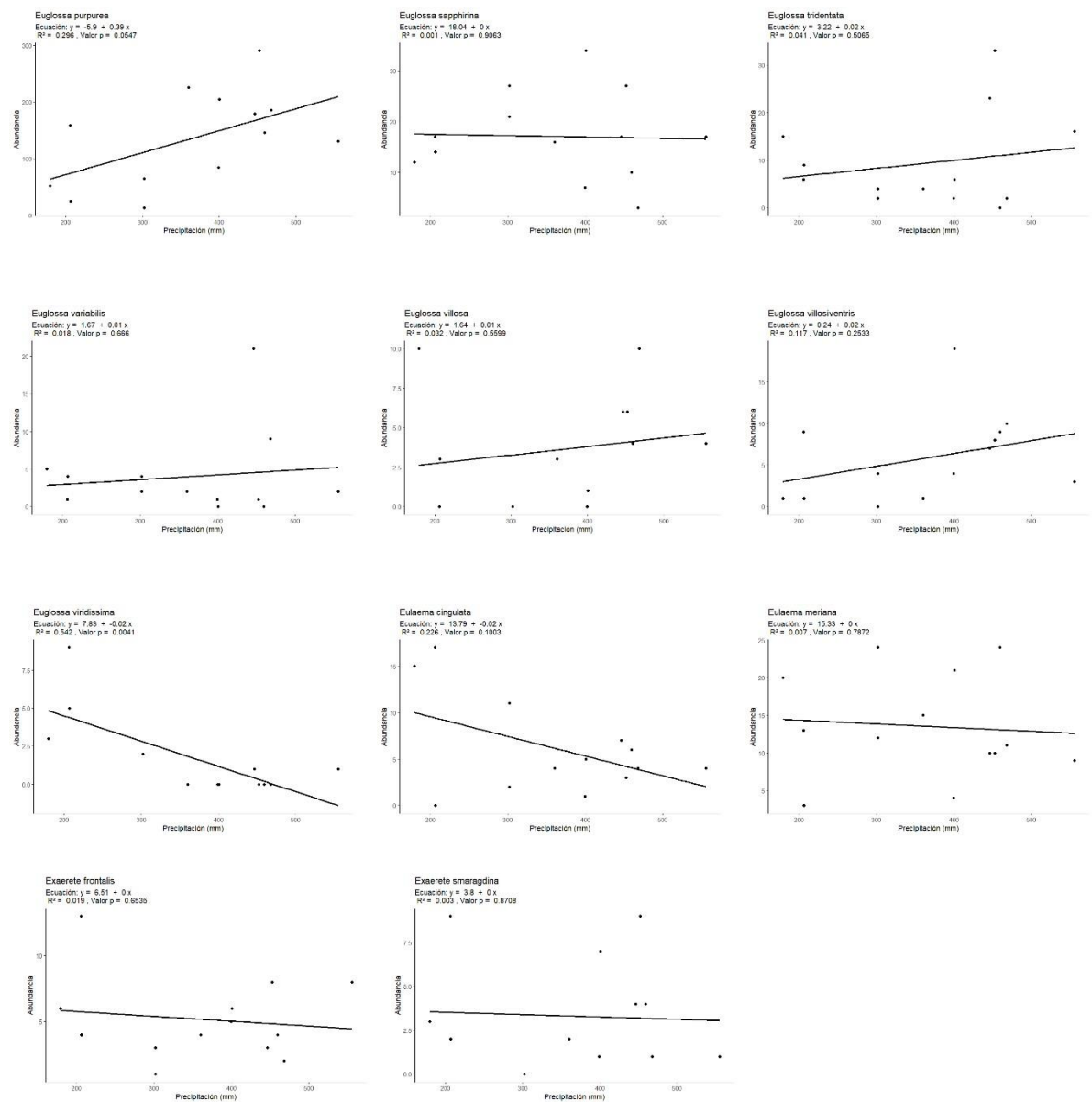


Figure. 5. Linear regression between abundance and precipitation of each species collected in the research ($n \geq 13$).

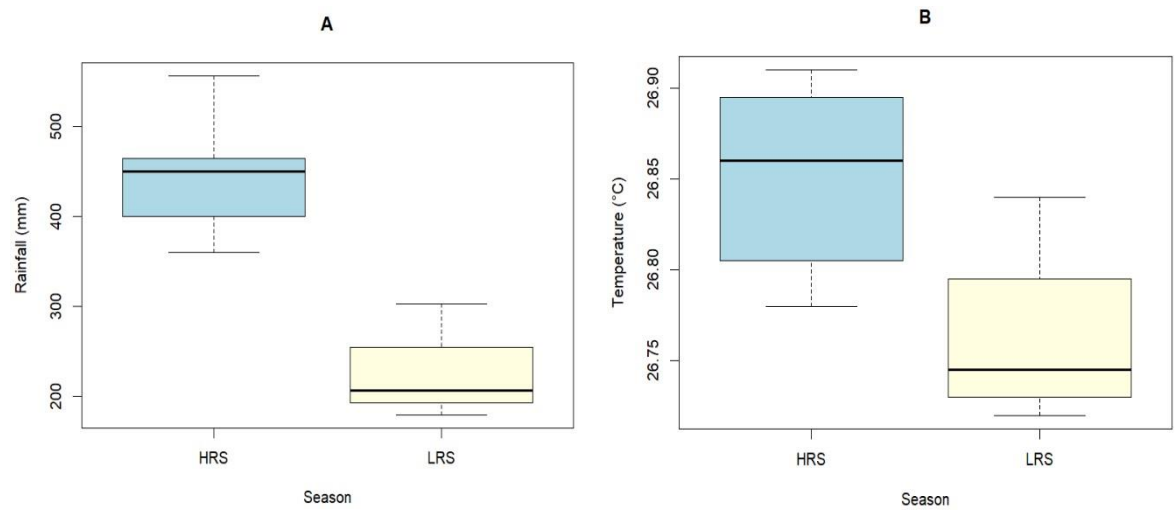
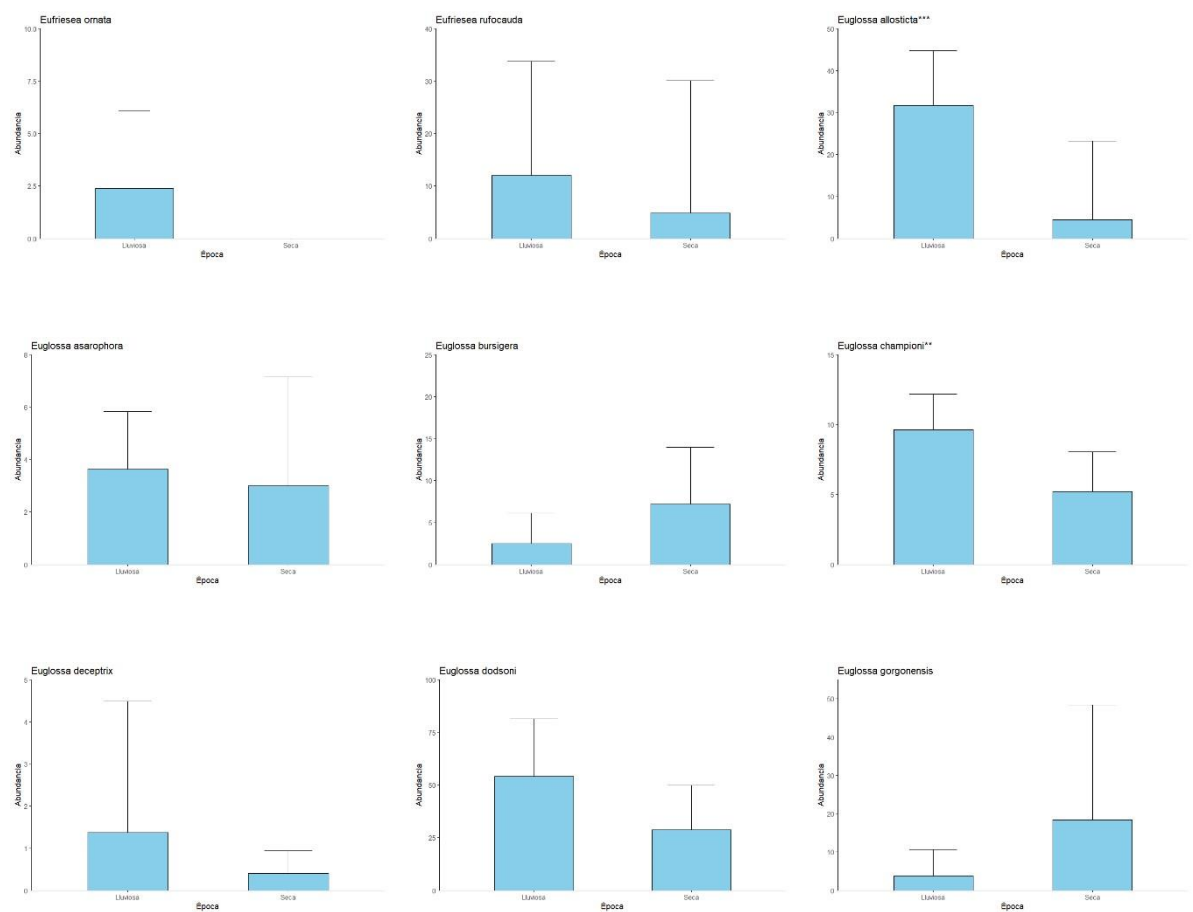


Figure 6. Average precipitation (A) and temperature (B) in each of the climate stations.



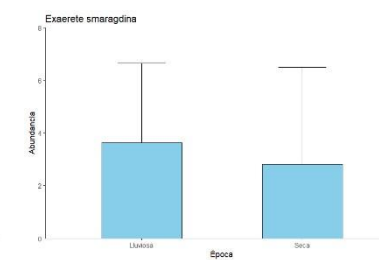
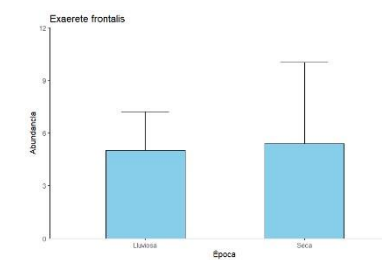
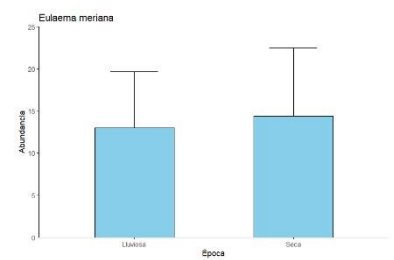
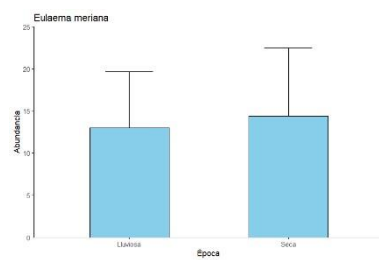
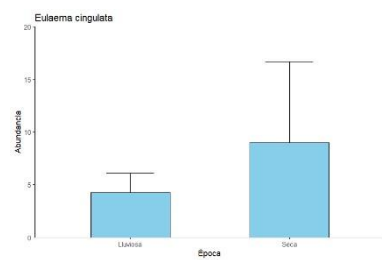
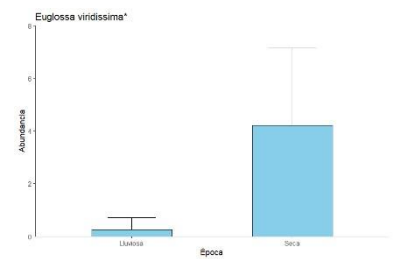
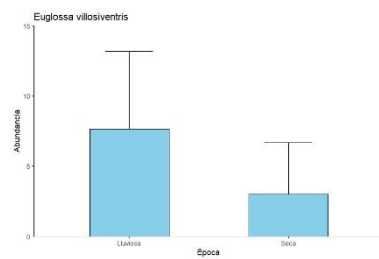
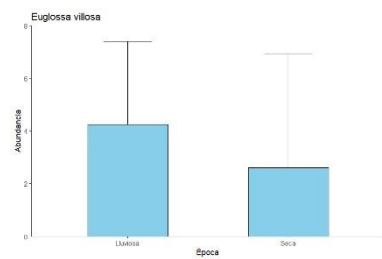
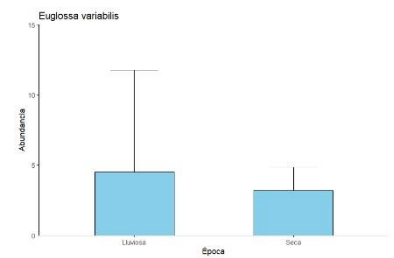
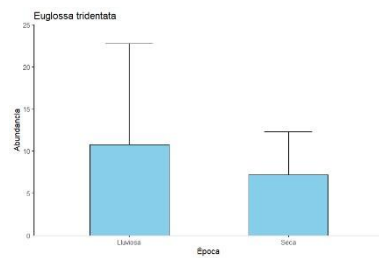
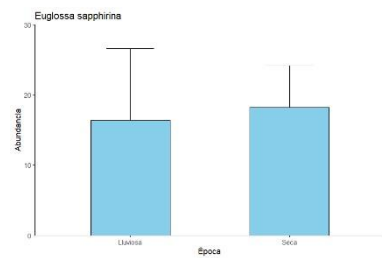
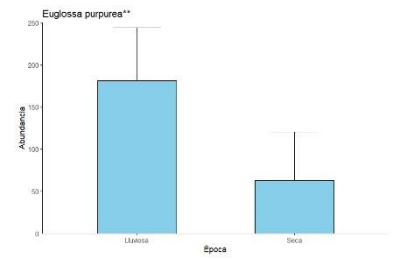
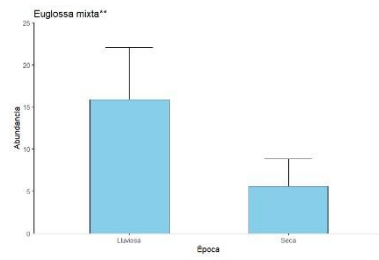
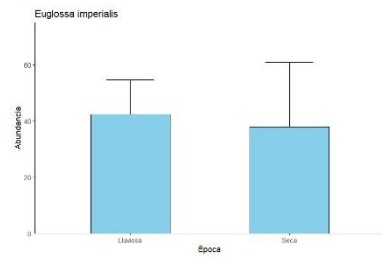
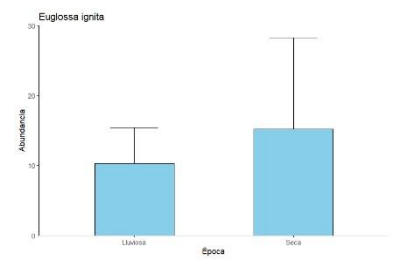
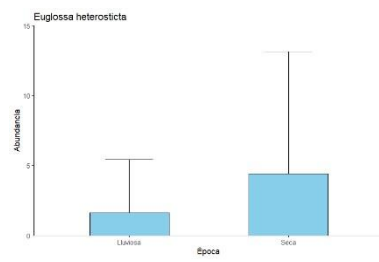
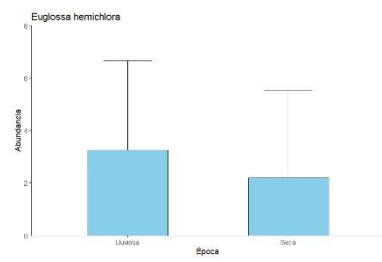


Figure 7. Results of t-student test for each species with respect to season (LRS and HRS). * $p > 0.05$. ** $p > 0.01$. *** $p > 0.001$

POLLINATION NETWORKS BETWEEN ORCHIDS AND ORCHID BEES: THE EFFECT OF PHENOLOGY ON MUTUALISTIC INTERACTIONS

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SUMMARY:

Pollination is a key interaction for biodiversity and ecosystems. In the neotropics, Euglossini bees play a crucial role in orchid pollination, but little is known about how climate changes, such as precipitation, affect these interactions. Although bees depend on flower resources, the dynamics of these relationships vary over time. This study analyzes how Euglossini bees and Orchidaceae interact in a humid tropical forest in Costa Rica and how precipitation influences those interactions. The study was conducted at the La Selva Biological Station, a tropical rainforest in Costa Rica, between February 2023 and January 2024. Male Euglossini bees were captured using six chemical attractants. The interactions between bees and orchids were analyzed through bipartite networks and various network metrics, such as specialization, nesting, and modularity. The effects of precipitation on the abundance of bees and pollinaria were also evaluated through linear regressions and network comparisons. A total of 643 pollinaria of orchids and 623 male Euglossini bees were collected, representing 25 species. *Gongora* spp. were the dominant species. In the dry season, 178 pollinaria from 17 species were collected, and in the rainy season, 465 pollinaria from 27 species were collected. A total of 96 interactions between bees and orchids were identified, revealing differences between seasons. The network during the rainy season was relatively large, but both stations had similar levels of nesting. There was a positive correlation between

males and pollinaria. This study is pioneering in documenting the phenology of the interactions between Euglossini bees and orchids in the Neotropical region, highlighting the impact of precipitation on these ecological networks. Unlike previous studies where interactions were concentrated in dry seasons, greater activity was observed during the rainy season. Future studies should integrate molecular tools and chemical analyses to improve species identification and better understand the relationships between these pollinators and orchids.

INTRODUCTION:

Pollination is one of the most important biotic interactions due to its essential role in plant reproduction and its contribution to the conservation and sustainability of biodiversity and global ecosystems (Stangler et al., 2016; Ollerton, 2017; Lacher et al., 2019; Stephens et al., 2023). This interaction is shaped by multiple factors, including the phenology of both pollinators and plants, as well as the morphology of flowers and pollinators, which affect how well they match. Additionally, the nested structure of interaction networks plays a key role in the stability and resilience of pollination systems, allowing for a better understanding of how these interactions behave in nature (Bascompte and Jordano, 2007; Elzinga et al., 2007; Nielsen and Bascompte, 2007; Ings et al., 2009; Encinas-Viso et al., 2012). Furthermore, the seasonality of the interacting species, influenced by factors like climate and resource availability, significantly affects both the occurrence and intensity of pollination interactions, as it determines when and how strongly these species interact (Morente-López et al., 2018; Ramos-Jiliberto et al., 2018)

Most phenological processes in plants are determined by climate patterns (Chmura et al., 2019; Piao et al., 2019). Similarly, the presence of pollinators depends largely on the climatic conditions (Vizentin-Bugoni et al. 2018; Bergamo et al. 2021). Bees play a fundamental role in the reproduction of plants because of their dependence on flower resources (Michener 1974, 2007; Roubik and Hanson 2004).

There are numerous examples that show a certain synchrony between the reproductive periods of the bees, and the plants they visit (López-Urbe et al. 2016; Dorchin et al. 2018). Phenological uncoupling in plant-bee interactions occur due to climate variation and the functional traits of plants or bees, where the periods of flowering, larval emergence, nest location, etc., influence the coupling and presence of the interaction (Robbirt et al., 2014; Danforth et al., 2019; Stemkovski et al., 2020). In the tropics, phenological-interaction studies have been conducted in ecosystems that are climatically seasonal, such as tropical dry forests (Flórez-Gómez et al., 2020; Cortés-

Flores et al., 2023). Phenological studies on bee–plant interactions generate information on the effects of time, climate change, the introduction of exotic species and the loss of habitat, in interactions at different times of the year (Dalsgaard et al. 2013; Morton and Rafferty 2017).

Euglossine bees (Apidae: Euglossini), known as "orchid bees", are among the most important pollinator groups in neotropical forests, where they constitute 25% of the diversity of bees in these ecosystems (Roubik and Ackerman 1987; Roubik and Hanson 2004; Nemésio and Silveira 2007; McCravy et al. 2016). Orchid bees are responsible for pollinating a wide diversity of plants, including at least 1,300 species of orchids (10% of Neotropical orchid diversity) (Ackerman et al., 2023). Male Euglossini bees collect fragrances from various sources, such as decaying organic matter, feces, and some orchids (Dressler 2003; Roubik and Hanson 2004). These fragrances are used for their reproduction, territorial defense and species recognition (Henske et al. 2023).

Male Euglossini bees pollinate flowers of orchids belonging to various lineages, including species in the Epidendroideae and Vanilloideae subfamilies. Among these we find the genera *Catasetum*, *Cycnoches*, and *Mormodes* in the subtribe Catasetinae; *Acineta*, *Coryanthes*, *Gongora*, and *Stanhopea* in the subtribe Stanhopeinae; *Macroclinium* and *Notylia* in the subtribe Oncidiinae; and *Cochleanthes*, *Dichaea*, and *Keferstenia* in the subtribe Zygopetalinae, as well as *Vanilla* in Vanillinae subtribe, among others (Ramírez et al. 2002; Watteyn et al. 2021). Most orchids use these chemical rewards to attract bees and increase the probability of reproduction, whereas others use nutritional deception strategies (Sabat and Ackerman, 1996; Jersáková et al., 2006; Karremans, 2023). Orchids that have benefited from this behavior have diversified rapidly (Ramírez et al. 2011).

Despite the importance of euglossine bees, studies have only superficially mentioned their interactions with orchids without considering the effects of temporal climate variation on these interactions, despite considerable sampling efforts (Roubik and Ackerman, 1987). Information on temporal variations in the interaction networks between pollinators and orchids over the year is lacking, which has led to the assumption that such interactions occur throughout the year, regardless of weather conditions. However, many orchids have short flowering periods (Pfeifer et al., 2006); thus, we hypothesized that the pollination interactions between orchids and orchid bees are dynamic and may be dependent on the reproductive phenology of the orchids and the bees.

Costa Rica is the Central American country with the highest diversity of orchids (~1,700 species) (Pupulin et al., 2023) and is first in number of Euglossini species per unit area (1 species/775 km²) (Ramírez et al., 2002; Roubik and Hanson, 2004). The ability to

attract male orchid bees loaded with orchid pollinia offers the opportunity to study this interaction in detail and to explore the effect of environmental changes on the presence or strength of this interactions. It also allows measuring the effect of bee abundance on this process. The possibility to identify pollinia with a high level of taxonomic resolution is the main limitation since orchids should be identified to the maximum level possible in order to register the precise interactions. This work aims to characterize the dynamics of the interactions between euglossine bees and orchids in a humid tropical forest in Costa Rica, as well as the influence of environmental variables and bee abundance on these dynamics.

METHODOLOGY:

Study area:

The study was carried out at the La Selva Biological Station (LSBS), owned by the Organization for Tropical Studies (OTS), in a tropical humid forest ecosystem (Holdridge, 1967) in Puerto Viejo, Sarapiquí, Heredia, Costa Rica, from January 2023 to January 2024. The LSBS has an average elevation of 35 m above sea level and a surface area of 1,536 hectares. It has a slightly bimodal rainfall regime of approximately 4,000 mm/year, with July and December being the months with the highest rainfall (> 500 mm and > 400 mm, respectively), while the driest months are February and March (<200 mm) (Sanford et al. 1994). The temperature is constant throughout the year (~ 25 °C). The LSBS is in the foothills of Braulio Carrillo National Park in the Central Volcanic Mountain Range of Costa Rica. Within the LSBS, three sampling points were located approximately 3km from each other. All sampling points were in mature forests and were approximately 40–50 years old.

Sampling of male orchid bees:

Fieldwork was performed monthly during similar dates, during three consecutive days (2 days for January 2023 only), each day corresponding to a different sampling point. Six chemical compounds were used to attract male orchid bees: skatole, vanillin, 1,4-dimethoxybenzene, eugenol, methyl salicylate and eucalyptol. Crystalline attractants were scattered on the ground and macerated with a stone; the liquid chemicals were placed on cotton balls held on branches 1.5 meters from the ground. Each bait was separated by at least 15 meters.

Each bait was monitored for 5 minutes every half hour. During this time, all euglossine bees with pollinia or accessory structures (viscidium, stipe or caudicle) on the body were captured. Each male was individually placed in a 1.5 mL Eppendorf® tube and placed in

a cooler with ice. Each male was then inspected with a magnifying lens to confirm the presence of the pollinia, tagged and the location of the pollinia was recorded. Finally, all the males captured with pollinia and/or accessory structures were frozen. The abundance of males without pollinia was determined as described in Chapter 1.

Identification of pollinia and orchid bees:

The identification of the pollinia was carried out at Lankester Botanical Garden (JBL, acronym in Spanish). Orchid pollinaria or their remnants, including accessory structures, were examined in detail under a stereo microscope and photographed in high resolution. We used the photographic equipment available in the CIJBL and the software Zerene Stacker version 1.04 and Adobe Photoshop 2024 to obtain photographs with the greatest possible detail.

For the identification of the pollinaria, we used the guides of Pupulín (1995, 2001, 2007, 2010), Pupulin & Karremans (2008) and Fernández et al., (2011), as well as the orchid pollinaria database and high-resolution photographs. Fresh pollinia from the JBL living orchid collection as well as the wet pollinia reference collection were used to confirm the morphological identification. A review of possible orchid pollinators was also carried out based on the complete list of known orchid pollinators available in supplementary files from Ackerman et al. (2023).

Bees were identified using the keys of Roubik and Hanson (2004) and Hinojosa-Díaz and Engel (2011, 2014). We performed comparisons with specimens in the Entomological Collection of the Museum of Zoology (MZUCR), of the Biodiversity and Tropical Ecology Research Center, University of Costa Rica. The bees were identified to the lowest possible taxonomic level and deposited in the Collection of Floral Visitors of the JBL. Reference specimens were also deposited at MZUCR.

Obtaining climate data:

The climate data were obtained from the OTS automatic climate station at LSBS. The data used for the analyses were the accumulated averages of the last 53 years. For the present work, two seasons were defined: the low rainy season (LRS), from January to April, with rainfall < 300 mm per month, and the high rainy season (HRS), from May to December, with rainfall > 300 mm per month.

Networks of interaction and statistical analysis:

A matrix was constructed for each climate station based on the number of euglossine males with pollinaria. Then, bipartite networks were constructed to illustrate the

interactions for each season. Network metrics were calculated via the *bipartite* package (Dormann et al., 2008); network graphics were generated via the *plotweb* and *circlize* packages. We used the database from Ackerman et al. (2023) to compare the interactions that had already been reported to identify possible new interactions.

Network metrics were calculated to compare the interactions between each rainy season. We used several network metrics: 1. Specialization index (H_2'), which ranges from 0 to 1, where 0 means that there is no specialization and 1 means that the network is completely specialized; 2. Nestedness (NODF), a metric that measures the degree to which specialized species interactions are subsets of more general species interactions in the network, ranging from 0 (not nested) to 100 (perfectly nested); 3. Connectivity, a metric used to calculate the observed proportion of all the possible interaction in the network; 4. Modularity, to determine the formation of modules or interactions within groups of orchids or bees; and 5. Uniformity of the interaction, the homogeneity in the frequencies of the interactions. We compared each metric with a null model proposed by Bascompte et al. (2003), which assumes the probability of an interaction as proportional to both communities. The null model is based on 1000 iterations, keeping the sum of rows/columns of the matrix constant. For this analysis, the null model function *r2dtable* was used in the *bipartite* package (Dormann et al., 2008). Observed metrics were significant when greater than 95% of the randomly generated metrics.

We estimated the dissimilarity of the species interaction networks (Poisot et al., 2012) as the rotation of interactions between plants and floral visitors, using the Whittaker dissimilarity index, which is calculated via the *betalinkr* function of the *bipartite* package in R. This index measures the degree of variation in the specific interactions between plants and flower visitors across the rainy seasons. With the *betalinkr* function, we broke down beta diversity (β_{WN}) into two additive components: composition rotation (β_{ST}) and rewiring of interactions (β_{OS}) between shared species. In addition, we calculated the relative contribution of the differences in the composition ($\beta_{ST}/\beta_{WN} \times 100$). This index varies from 0 to 100%, where values of $\beta_{ST}/\beta_{WN} > 50\%$ indicate high turnover, reflecting large differences in the composition of species and interactions between networks. Values of $\beta_{ST}/\beta_{WN} < 50\%$ suggest a lower degree of species rotation and a greater influence of rewiring on beta diversity. For all comparisons, we applied the “*commondenom*” method to decompose the β diversity components between the networks. All network metrics were calculated via the 'bipartite' package in the R statistical language.

We used Euglossini male abundance data (Chapter 1) and performed linear regressions to determine whether precipitation influenced the abundance of males, the abundance

and diversity of pollinaria, and the relationship between the abundance of males and the abundance and diversity of pollinaria.

RESULTS:

A total of 643 pollinaria or their remnants, including accessory structures, belonging to 2 subfamilies, 3 tribes and 5 subtribes, 17 genera and 34 orchid morphospecies were collected. The months with the greatest number of pollinaria present on Euglossini bees were June and August, while the month with the lowest number was September (**Table 1**). *Gongora* Ruiz & Pav., was the dominant genus throughout the year, with abundance peaks in June and July (**Table 2**).

The LSBS has a slightly bimodal precipitation regime, and we decided to unify the months of the high rainfall season (HRS) (>300 mm) and the low rainfall season (LRS) (<300 mm) for the data analysis. A total of 178 pollinaria belonging to 17 species were found in the LRS, while 465 pollinaria belonging to 27 species were found in the HRS. *Gongora cassidea* Reichenbach f., *Houlletia tigrine* Linden ex Lindl. & Paxton, *Macradenia brassavolae* Reichenbach, and *Sobralia decora* Bateman were unique to the LRS, whereas pollinaria unique to the HRS included *Catasetum maculatum* Kunth, *Cynoches* sp. Lindl, *Dichaea acrolepara* Schltr., *Dichaea sarapiquensis* Mora & García, *Dressleria dilecta* Dodson, *Huntleya buriti* Endrés & Rchb., *Kefersteinia lactea* Schltr., and *Vanilla* sp. Mill. Four genera presented the highest abundance of pollinaria, where the majority had a peak abundance in June, except for *Sievekingia* Rchb. F., which had a peak abundance in January (**Figure 3**)

In total, the network presented 96 interactions, that is, 12% of the total possible interactions. For the LRS, the network presented 51 interactions out of 391 possible (13% connectance) (**Figure 1A**), whereas the HRS presented 78 interactions out of 648 possible (12% connectance) (**Figure 1B**). *Gongora* was the orchid genus that showed the most interactions, with a total of 18. *Euglossa purpurea* was the bee species that had the most interactions, with pollinaria of 13 different orchid species belonging to 10 genera in 5 subtribes, followed by *Euglossa dodsoni* and *Euglossa tridentata*, each with pollinaria from 10 different orchid species belonging to 8 genera in 5 subtribes.

We found that the networks had structural differences between the LRS and HRS seasons (**Table 4**). The size of the network, the number of links and the weight NODF were greater for the HRS than for the LRS. The network of the HRS was larger, but together, both networks had similar nestedness values (NODF and WNODF) (**Table 2**). The modularity and specialization of the networks also had very close values, with the

LRS having slightly higher values. Similarly, the connectance and uniformity of the interaction did not significantly differ between seasons. Similarly, the differences in the networks between seasons were explained much better by a rearrangement than by a replacement of species within the seasons (β ST/ β WN <50%). We found that there was a significant positive correlation between the abundance of male bees and the number of pollinaria collected (**Figure 2**).

DISCUSSION:

Euglossine bees are essential pollinators in neotropical forests. Their wide flight range favors their dispersal and the pollination of orchid species with low densities (Janzen, 1971). In this study we found pollinia of rare orchid species like *Dichaea acrolephara*, *Dichaea sarapiquensis* and even *Trichocentrum costaricensis*, whose pollinators had never been observed before. We also detected 25 new pollination interactions in this community, demonstrating that this indirect methodology is efficient in registering obscure interactions between epiphytic orchids and their pollinators. Canopy plants are highly diverse but due to the difficulty in accessing them, their pollinator interactions are poorly known and remain a frontier in pollination ecology studies (Blüthgen and Klein, 2011).

Interaction network diversity:

Bee–orchid interactions have been previously documented in La Gamba Field Station, Costa Rica (Ramírez 2019). This site is in a successional forest with a more pronounced dry season than La Selva. Pollinaria were found on all euglossine genera except *Eufriesea* even though the sampling effort was greater (5 years) than that of the present study. In both La Gamba and La Selva, *Gongora* were the dominant species in the samples. The diversity of orchids in La Gamba could be much greater than that in La Selva, and interactions with orchids were found that we did not find in the present study (*Stanhopea* Frost ex Hook., *Polycycnis* Rchb. F., *Lycaste* Lindl., *Trichopilia* Lindl., *Macroclinium* Barb. Rodr., and *Clowesia* Lindl.) (Ramírez 2019). In our work the taxonomic resolution was greater, identifying most of the orchids to species level. In addition, we found genera and species that were not found in La Gamba (*Macradenia* Rchb. F., *Trichocentrum* Poepp. & Endl., *Houlletia* Lindl. & Paxton, *Paphinia* Lindl., *Rodriguezia* Ruiz & Pav, *Huntleya* Bateman ex Lindl.).

Euglossa imperialis was the dominant bee species in the studies done at La Gamba (Ramírez, 2019) and Barro Colorado Island in Panama (Ackerman, 1983), while in our study it was *Euglossa purpurea*. We found species that were not reported in La Gamba, such as *Euglossa ignita*, *Euglossa allosticta*, and *Euglossa townsendi*. Euglossine

species richness was similar at the two Costa Rican sites (24 species in La Gamba, 25 species in La Selva), but greater at Barro Colorado (44 species), making the latter one of the best documented places and the one with the greatest diversity of euglossines.

Interaction networks phenology:

The networks in La Selva showed similar levels of nesting and connectivity, patterns that are repeated throughout mutualistic interaction networks (Elzinga et al., 2007; Nielsen and Bascompte, 2007; Morente-López et al., 2018; Ramírez, 2019; Flórez-Gómez et al., 2020; Cortés-Flores et al., 2023). The changes in the interaction networks between the low and high rainfall seasons may be primarily driven by variations in light availability (photoperiod) and rainfall, which are crucial for plant flowering. These changes are most noticeable during the transition months, when a higher level of plant flowering is expected (Sahagun Godinez, 1996; Zimmerman et al., 2007; Ramírez et al., 2015).

With more species of plants producing flowers, an increase in the abundance and diversity of Euglossini bees is expected due to the greater availability and diversity of resources (Potts et al. 2003; Blaauw and Isaacs 2014; Häussler et al. 2017). However, our results contrast with previous works carried out with pollination networks, where most of the interactions occur in times of lower rainfall; in our work, they occurred during the season of higher rainfall. In tropical humid forests in general, rain is the environmental factor that can induce the flowering of many species, including orchids (Potts et al. 2003). It is expected that changes in the availability of flower resources influence plant/pollinator interactions (Fontúrbel et al., 2017). Documentation of interaction network phenology processes is essential, as it provides valuable information on the dynamics of the relationships between both interacting groups and how environmental factors, especially precipitation, influence changes in the structure and composition of these ecological networks.

The networks presented here have very similar nesting values, which occurred when a small group of pollinators, in this case orchid bees, interacted with the same orchid species as generalist bees did (Nielsen and Bascompte 2007; Bascompte and Jordano 2007a). It is important to highlight the difficulty of working with genera such as *Gongora*, *Mormodes*, and *Coryanthes*, where morphological identification based on pollinia morphology is not possible for most species. We recommend that future studies integrate molecular tools to accurately identify plant species, such as those visited by *Euglossa purpurea*, which are dominant throughout the year and associated with 11 orchid morphospecies. In addition, chemical and functional trait analysis of bees and orchids

would contribute to a better understanding of the relationships between different orchids and euglossine bees.

Among the interactions observed in this work, many of the euglossine bees share pollinia of the same genus or the same species of orchid, which suggests an evolutionary convergence of orchids to use the strategy of producing volatiles to attract males (Ramírez et al. 2015), but this hypothesis needs further study. However, the new interactions that occur during different seasons may be due to males changing their scent preferences, a phenomenon that has been reported in the literature (Abrahamczyk et al. 2012; Margatto et al. 2019; Darragh et al. 2023). These changes in scent preferences could be the consequence of changes in the availability of resources when a limited number of plants are producing them (Abrahamczyk et al. 2012).

Pollinaria-based orchid phenology:

Orchids that are pollinated by male euglossines have developed adaptations to increase the probability of flower visitation by generating volatile compounds that male bees require for courting females. Moreover, orchids avoid self-fertilization by producing few flowers during short periods thereby forcing the bees to travel great distances (Karremans 2023). For example, some species of *Dichaea* in Brazil have flowering periods that can vary from 4-60 days, during which time the pollinator community can vary (Nunes et al. 2017). However, most studies that have focused on the phenology of orchids visited by euglossine bees have been conducted in controlled environments. For example, *Gongora quinquinervis* was studied in Pernambuco, where they found that this plant had several flowering peaks throughout the year; the flowers remained open for several days (1-8) and most visits were by *Euglossa cordata* (Martini et al., 2003). *Gongora maculata*, on the other hand, can have blooms of only 2 days (Dodson and Frymire, 1961). Similarly, some genera such as *Stanhopea* and *Catasetum* have very short flowering periods, due to the high energy cost of producing inflorescences (Gerlach and Schill, 1991). For the other genera found in the present study there is no information available on flowering periods. Finally, it has been suggested the interactions have an intrinsic phylogenetic component, whereby the most recent lineages of bees are expected to have preferences for the most recent lineages of orchids (Baguette et al., 2020).

This is the first study in the Neotropical region to provide detailed information on the phenology of interactions between Euglossini bees and orchids, highlighting how environmental factors, especially precipitation, influence these ecological networks. Unlike previous studies, where plant–pollinator interactions tend to be concentrated

during the dry season, our results reveal high activity during the season of higher rainfall, which highlights the importance of precipitation in the induction of flowering in many orchid species. Integrating molecular tools, chemical analysis and functional traits can provide a solid framework for future studies that seek to understand in greater depth the interactions between orchid bees and orchids.

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SUPPLEMENTARY MATERIAL

Table 1. Orchid and bee richness and abundance at the La Selva Biological Station in 2023-2024

Month	Total pollinaria	Orchid species	Euglossine species
<i>Jan-23</i>	7	3	5
<i>Feb-23</i>	45	7	12
<i>Mar-23</i>	60	9	14
<i>Apr-23</i>	43	9	15
<i>May-23</i>	47	6	8
<i>Jun-23</i>	112	10	16
<i>Jul-23</i>	84	11	12
<i>Aug-23</i>	104	11	13
<i>Sep-23</i>	14	7	6
<i>Oct-23</i>	26	6	7
<i>Nov-23</i>	35	4	5
<i>Dec-23</i>	58	10	11
<i>Jan-24</i>	17	5	5

Table 2. Monthly abundances of each orchid species in LSBS

<i>Orchid species</i>	<i>January</i>	<i>February</i>	<i>March</i>	<i>April</i>	<i>May</i>	<i>June</i>	<i>July</i>	<i>August</i>	<i>September</i>	<i>October</i>	<i>November</i>	<i>December</i>	<i>Total</i>
<i>aff. Dichaea</i>		1											1
<i>Catasetum</i> sp.								2				1	3
<i>Coryanthes</i> sp.			2	4		5	3	4	3			1	22
<i>Cychnoches</i> sp.						5	2	5	1	1			14
<i>Dichaea</i> sp.	1	5	5	5	6	35	19	9		4	9	15	113
<i>Dichaea</i> <i>sarapiquensis</i>											1		1
<i>Dicheae</i> <i>acroblephara</i>												1	1
<i>Dressleria dilecta</i>							1			1			2
<i>Gongora cassidea</i>		1											1
<i>Gongora</i> sp.	13	22	39	25	30	56	49	26	5	5	14	20	304
<i>Gongora</i> sp.1									1				1
<i>Houlletia</i> sp.			2										2

<i>Huntleya burtii</i>									1				1
<i>Undeterminated</i>	1	2					1	1	2	2		2	11
<i>Kefersteinia aff. excentrica</i>			1			1		1					3
<i>Kefersteinia costarricensis</i>					1	1							2
<i>Kefersteinia lactea</i>											1		1
<i>Kefersteinia parvilabris</i>			2	1	1	5	2	1					12
<i>Kefersteinia sp.</i>						2							2
<i>Kegeliella kupperi</i>												2	2
<i>Macradenia brassavolae</i>	1												1
<i>Mormodes sp.</i>	3	8	1	2	7		1					1	23
<i>Notylia sp.</i>									1				1
<i>Paphinia subclausa</i>						1							1

<i>Rodriguezia compacta</i>	3							51		9		3	66
<i>Sievekingia fimbriata</i>		5	3	1	2	1		2					14
<i>Sievekingia suavis</i>				3			1						4
<i>Sobralia decora</i>				1									1
<i>Sobralia</i> sp.			5	1			3						9
<i>Trichocentrum costarricensis</i>	1											1	2
<i>Vanilla</i> sp.							1						1
<i>Zygopetalinae</i> sp.								1					1
Total	23	44	60	43	47	112	83	103	14	22	25	47	643

Table 3. New reports of interactions between orchids and euglossine bees found at the La Selva Biological Station, Costa Rica. Eg: *Euglossa*, Ef: *Eufriesea*, El: *Eulaema*, Ex: *Exaerete*

Orchid species	Euglossine bees
<i>Rodriguezia compacta</i>	<i>Eg. cybelia</i> , <i>Eg. dodsoni</i> , <i>Eg. purpurea</i> , <i>Eg. tridentata</i> and <i>Eg. dilemma</i>
<i>Dichaea sarapiquensis</i>	<i>Eg. dodsoni</i> , <i>Eg. erythrochlora</i> , <i>Eg. hemichlora</i> , <i>Eg. ignita</i> , <i>Eg. imperialis</i> , <i>Eg. purpurea</i> , <i>Eg. variabilis</i> and <i>El. meriana</i>
<i>Dichaea globosa</i>	<i>Eg. mixta</i>
<i>Dressleria dilecta</i>	<i>Eg. dodsoni</i> , <i>Eg. purpurea</i>
<i>Kefersteinia aff. excentrica</i>	<i>Eg. hansonii</i> , <i>Eg. purpurea</i>
<i>Kefersteinia lactea</i>	<i>Eg. purpurea</i>
<i>Kefersteinia costarricensis</i>	<i>Eg. purpurea</i>
<i>Kefersteinia parvilabris</i>	<i>Ex frontalis</i>
<i>Kegeliella kuperi</i>	<i>Eg. dodsoni</i> , <i>Eg. heterosticta</i>
<i>Macradenia brassavolae</i>	<i>Eg. tridentata</i>

Table 4. Metrics of the interaction of mutualistic networks for the LRS and HRS at the La Selva Biological Station. The estimated network metrics were compared with those of randomly generated models (N = 1000). Values in bold indicate a significant difference from random expectations ($p < 0.01$ for all cases).

<i>Network metrics</i>	LRS	HRS
<i>Species of bees</i>	23	24
<i>Orchid species</i>	16	27
<i>Links</i>	357	648
<i>Frequency of interactions</i>	0.44	0.58
<i>Nesting (NODF)</i>	35.87	30.26
<i>Degree of nesting (WNODF)</i>	22.73	19.54
<i>Modularity (Q)</i>	0.34	0.31
<i>Specialization (H_2')</i>	0.40	0.32
<i>Connectivity</i>	0.13	0.12
<i>Interaction level</i>	0.54	0.49
<i>Generalization (visitors)</i>	5.6	4.18
<i>Vulnerability (plants)</i>	2.52	3.34

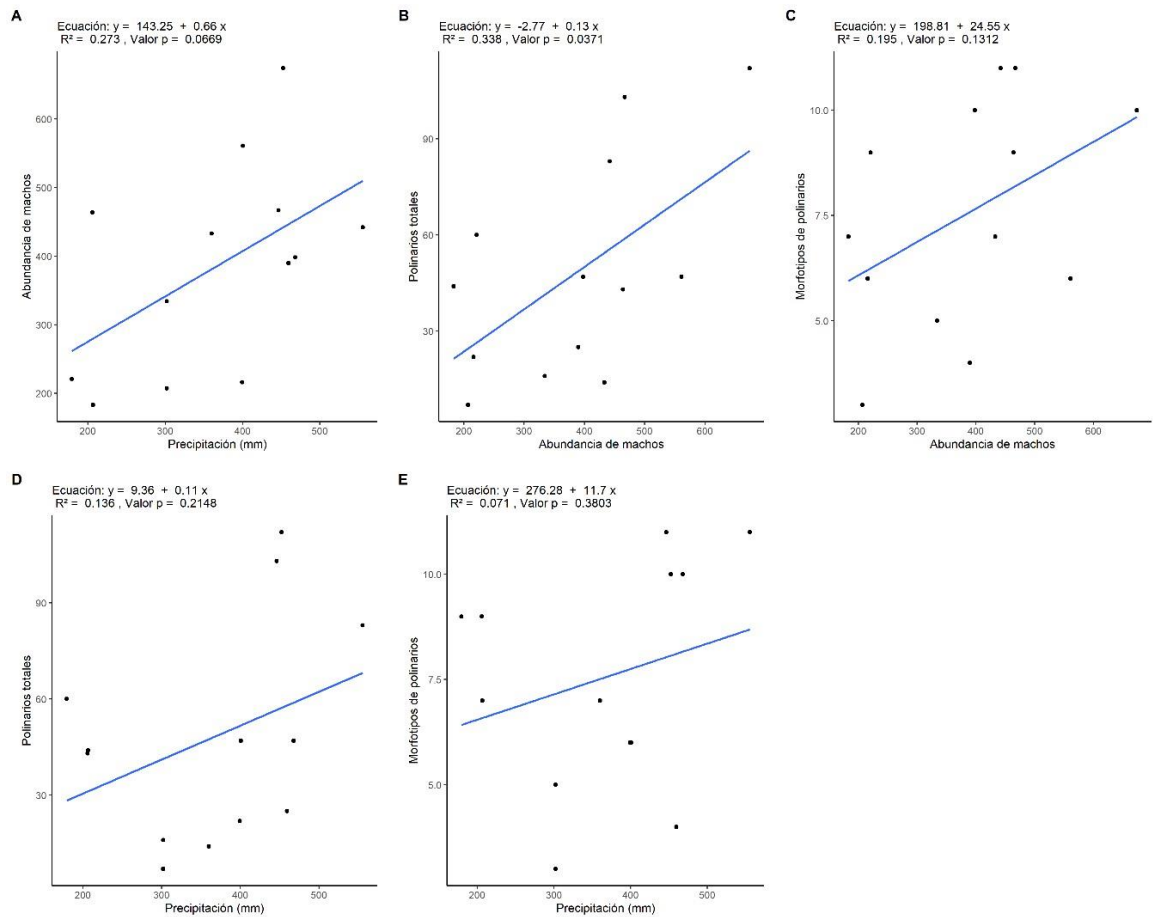


Figure 2. Linear regression between (A) precipitation and abundance of males, (B) abundance of males and total pollinaria, (C) abundance of males and pollinaria richness, (D) precipitation and total pollinaria, (E) precipitation and richness of pollinaria.

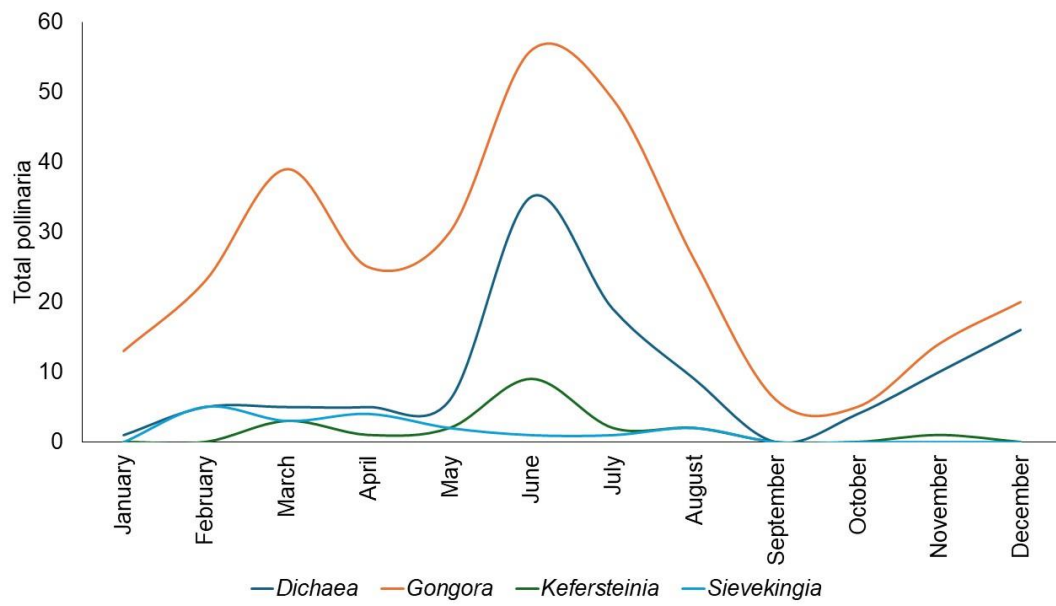


Figure 3. Phenology of the most abundant orchid genera in La Selva Biological Station during 2023-2024.